

World Models for Human Behavior

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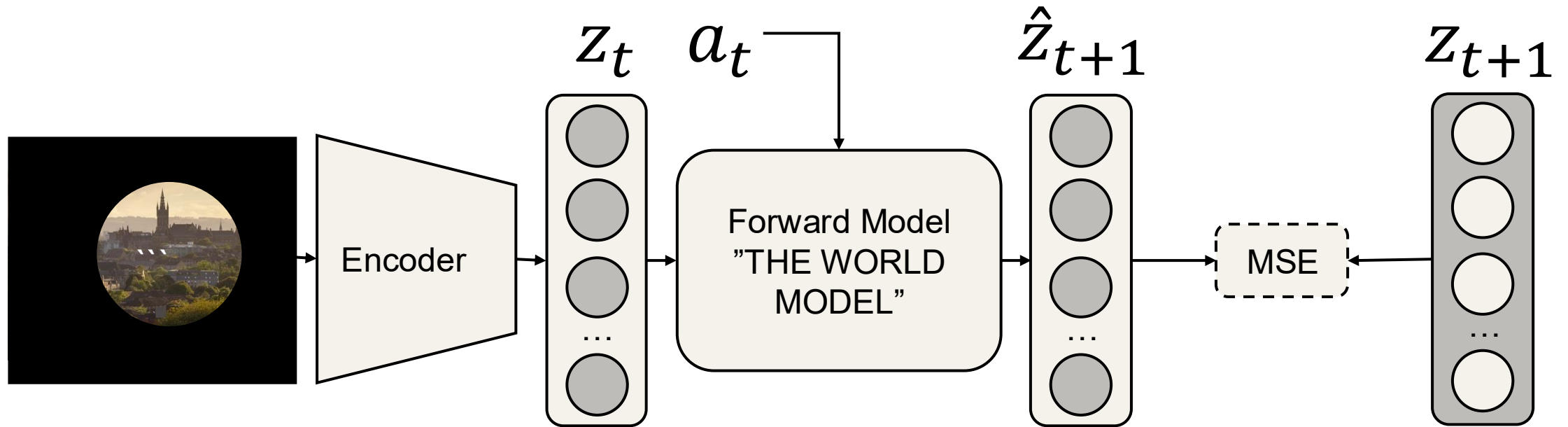
WORKSHOP ON THEORY IN COMPUTATIONAL INTERACTION

Example: Where to return?



1. Compress the visual information (encode)
2. Make a plan based on weight shape (plan)
3. Execute (act)
4. Reevaluate (encode $\rightarrow$ plan $\rightarrow$ act)

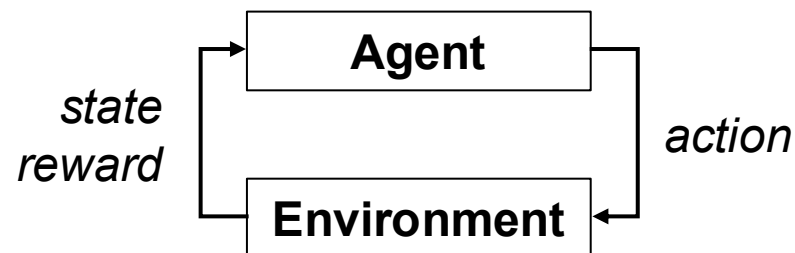
World Models



A world model learns to predict what the world will look like next, in a compressed latent space, given the action you take, and it trains itself purely on how wrong that prediction was.

Orthogonal to current Computational Rationality research

- Current user models use computational rationality
 - Resources are bounded, human behavior is approximately optimal within those bounds.
- In practice:
 - Model-Free Reinforcement Learning, with some noise in the action space and partial observability.



In model-free RL we try to learn a policy (a function that maps state to actions), that maximized future discounted reward.

Model-Free RL as backbone for user models is not sufficient.

- 1) Cannot predict the world
- 2) Cannot explicitly plan over a horizon
- 3) Does not generalize, no zero-shot capabilities.
- 4) Does not allow for bounds on planning itself.
- 5) Sometimes Model-free RL does not reach human performance, how should we ever build a CR model for this task? Example: Atari (not even too complex)

Why do we still train babies?

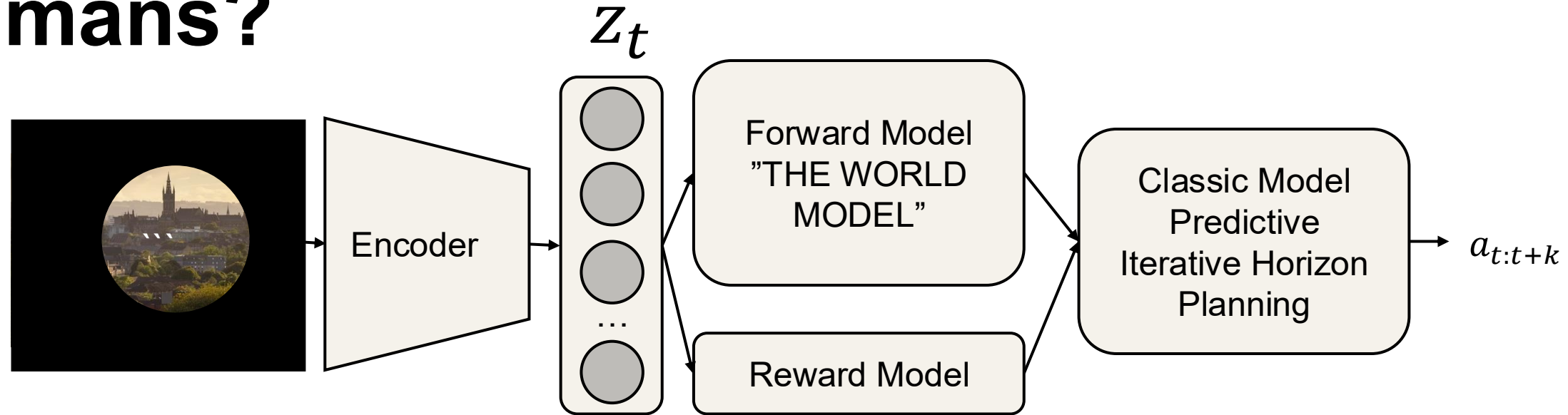
- Model-free RL



- Model-Based RL



Q1: How does the theory represent humans?



Assumptions:

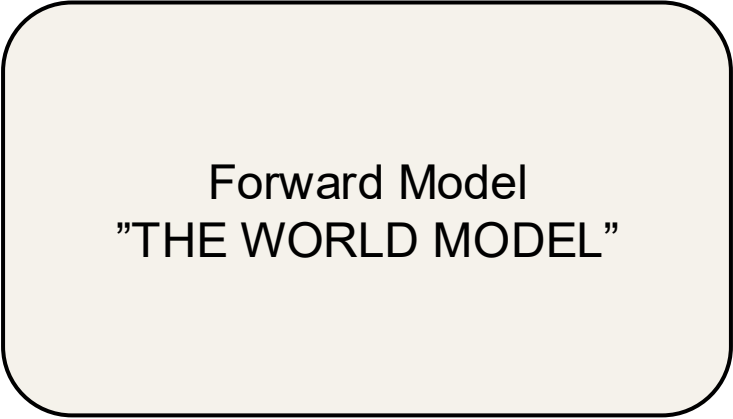
- Behavior is model-based not reflexive, planning runs in latent space,
- The model is learned self-supervised from prediction error, and it operates under finite depth, noise, and partial observation.

Psychology Perspective:

- There is evidence humans use predictive models of the world to plan a series of actions
 - E.g., perceptual neurons fire before the saccade happens.

Q2: How does the theory represent the computer or AI?

- The computer is part of the “world model”
 - Represents the human’s current understanding of how the system behaves.
- It is just transformations in latent space.



Forward Model
"THE WORLD MODEL"

Q3: What mechanisms link the two (human and computer)?

- See slide before.

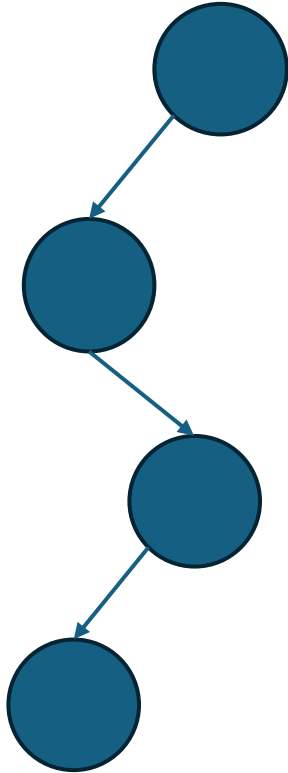
Q4: Which computational problems in interaction can be rigorously defined using the theory?

New opportunities

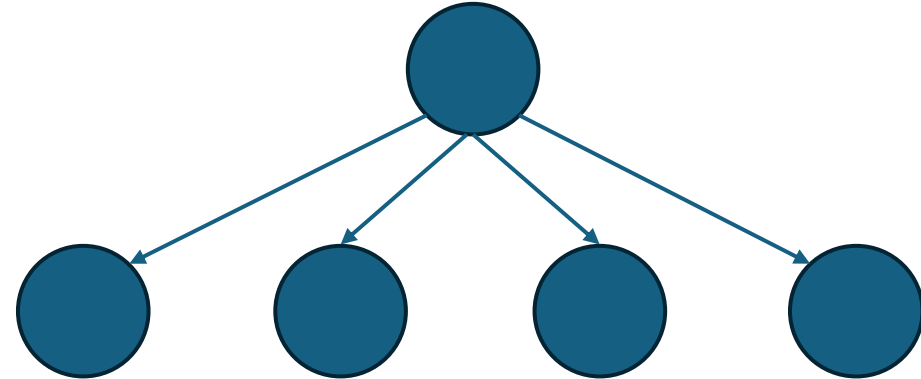
1. Unseen tasks
 - For different tasks in the same environment the model does not require retraining: One or few-shot adaption
2. Model Inversion
 - We could invert the model and ask why behavior happened.
3. Planning Limits
 - We could uniquely add more than just random sampling noise to the planning.
4. Flexible adjustment of behavior when goals change
 - With model-based approach adjusting to different goals within the same interface should be easier/possible

Model-based Planning as a theory

“Thinker”



“Creative”



Open Challenges

- No stable baselines3
- Latent spaces are not interpretable so harder to work with.
- Increased compute
 - Both training and inference
- But also opportunities: probably hierarchical with model-free RL
 - world models for decision-making and planning, RL for low-level behavior (movement, vision, etc.)
 - Can possibly interact with language based conditioning.